IY4113 Milestone 2

Assessment Details	Please Complete All Details
Group	B
Module Title	Applied Software Engineering using Object Orientated Programming
Assessment Type	Java Fundatmentals - Practical Part 1
Module Tutor Name	Jonathan Shore
Student ID Number	P0460817
Date of Submission	01/02/2026
Word Count	1200

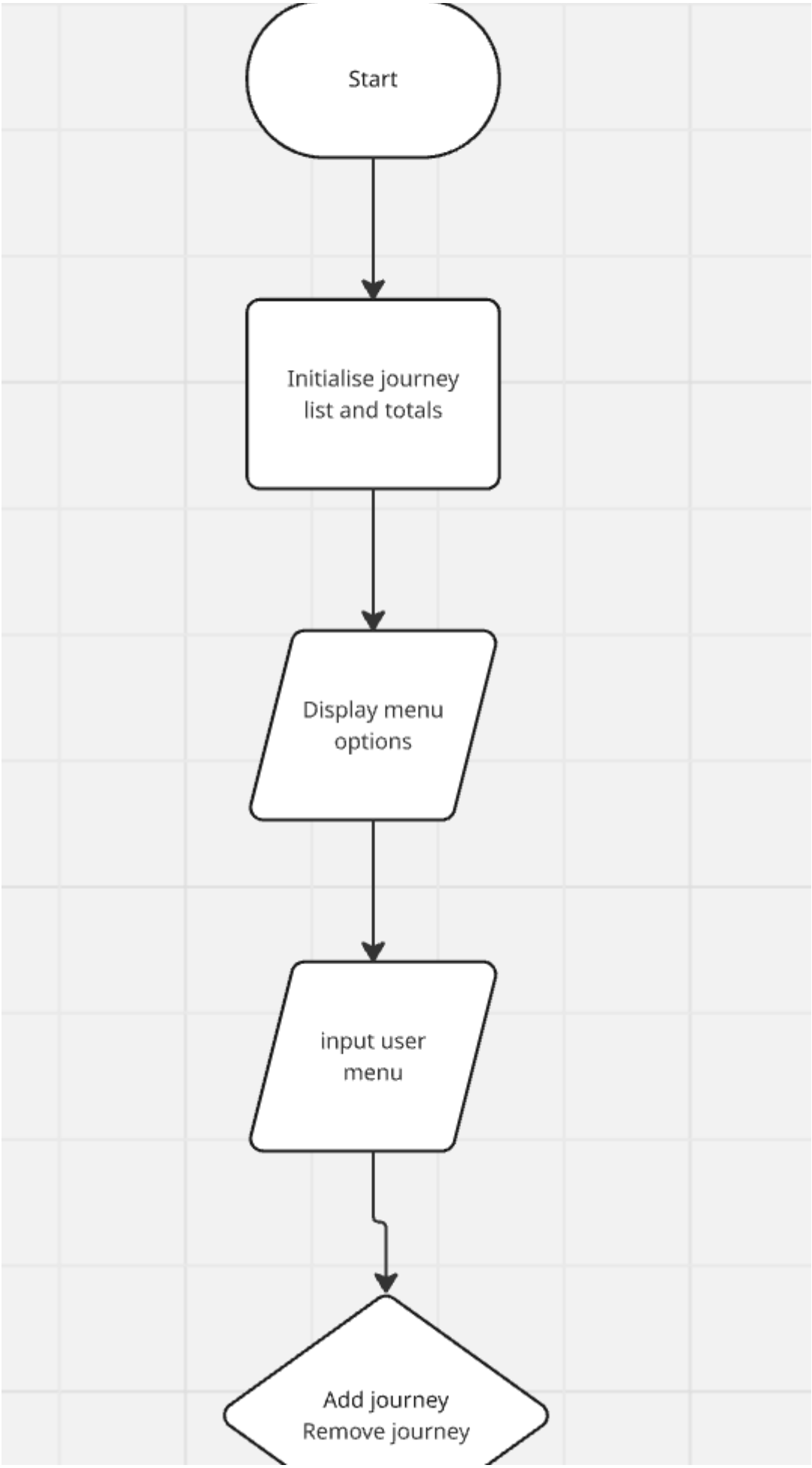
- [x] *I confirm that this assignment is my own work. Where I have referred to academic sources, I have provided in-text citations and included the sources in the final reference list.*
- [x] *Where I have used AI, I have cited and referenced appropriately.

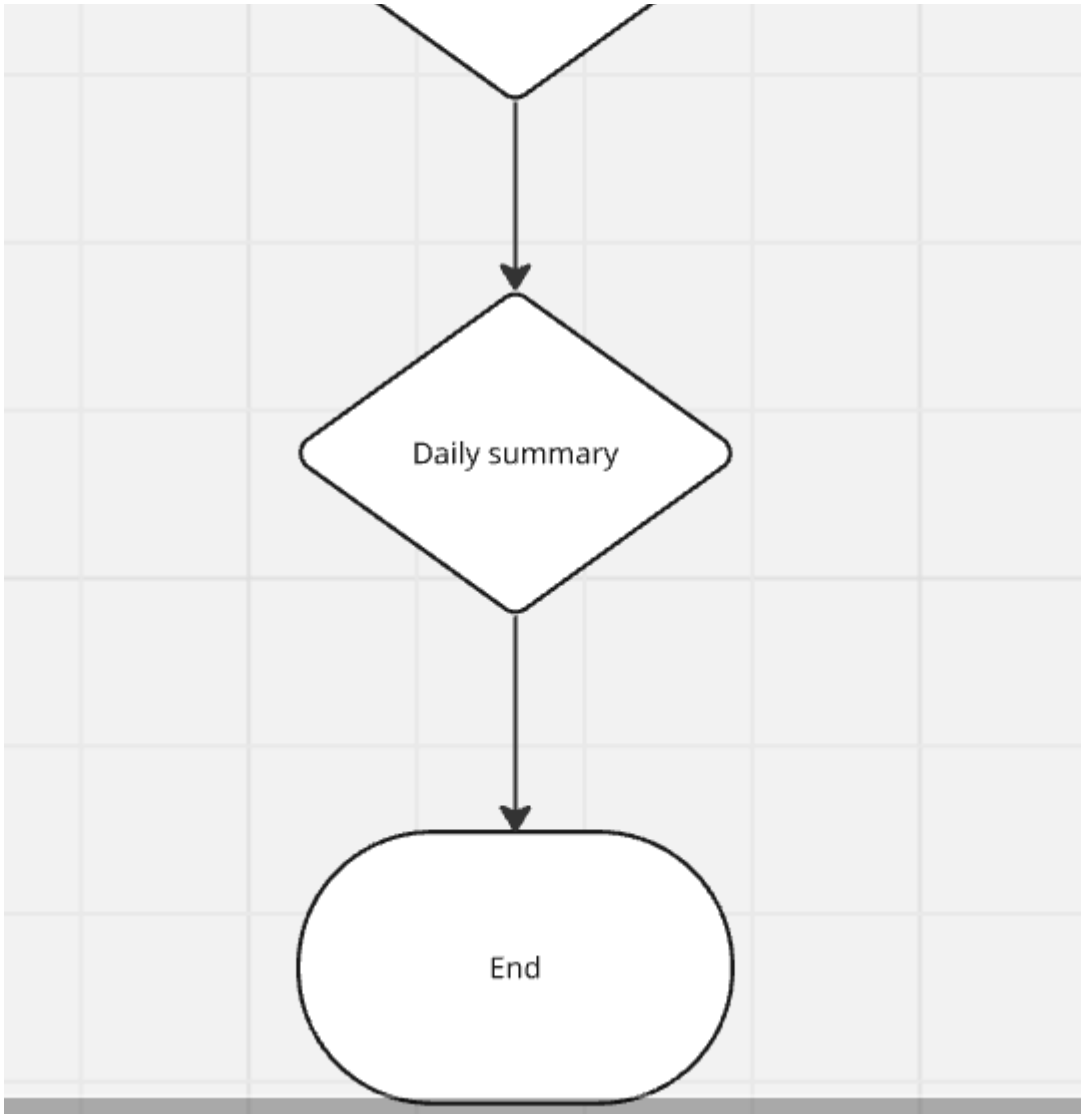
Algorithm Design

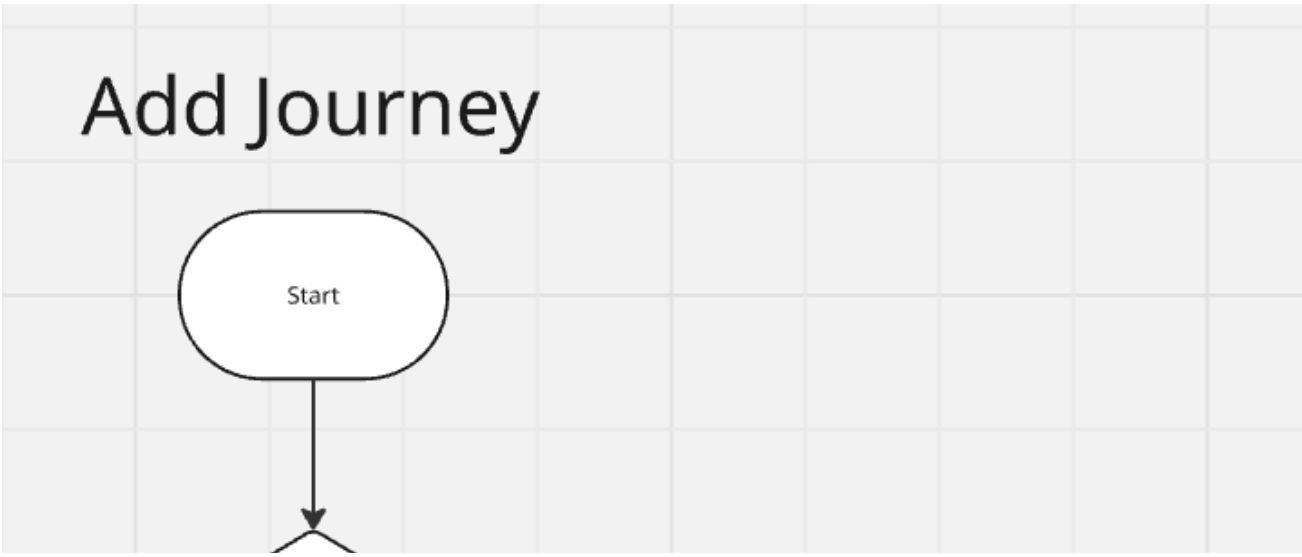
The algorithm design for the CityRide Lite program is explained using flowcharts and a class diagram.

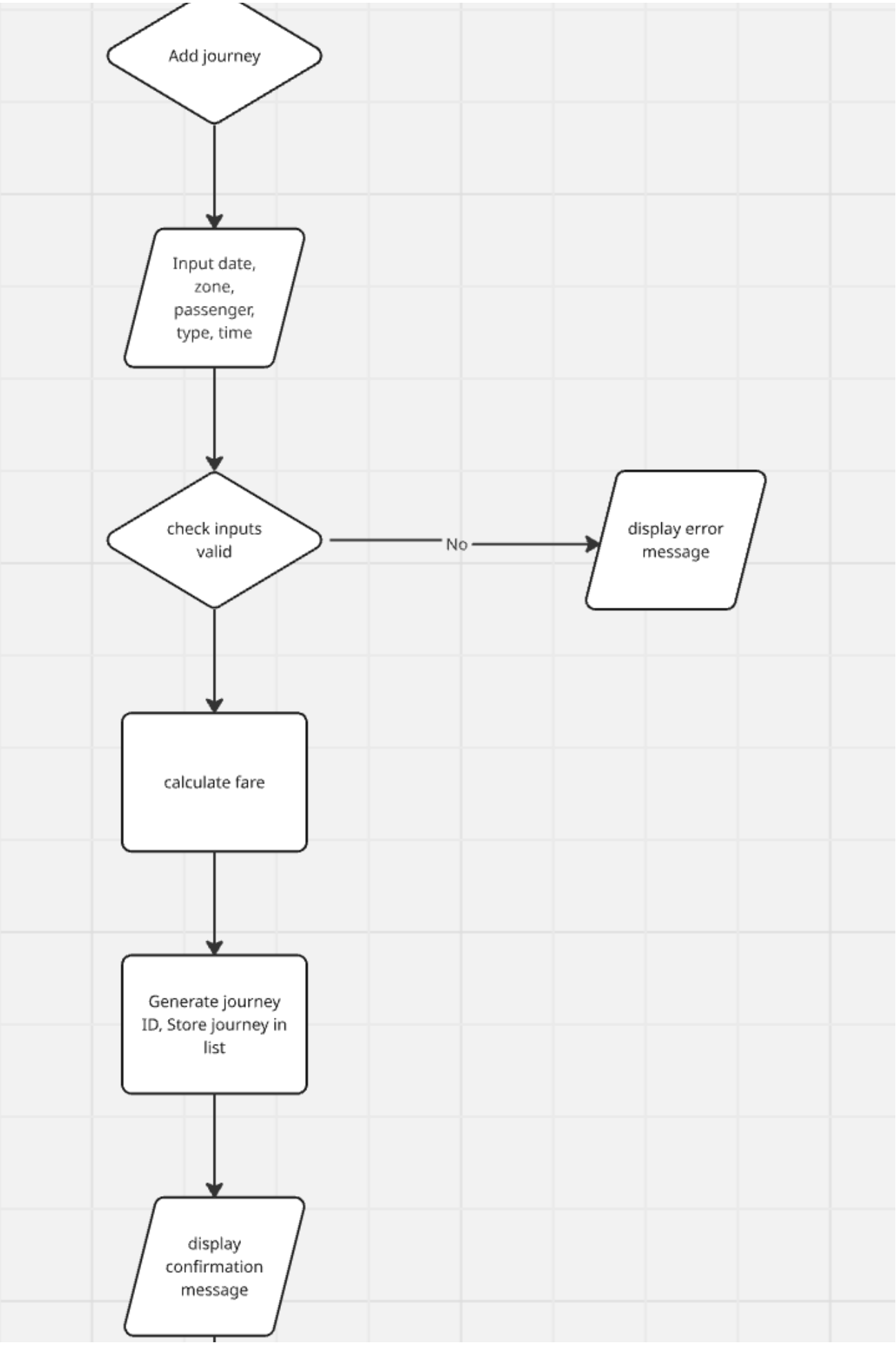
The program is split into different sections such as Main Menu, Add Journey, Remove Journey, Daily Summary and List Journeys.

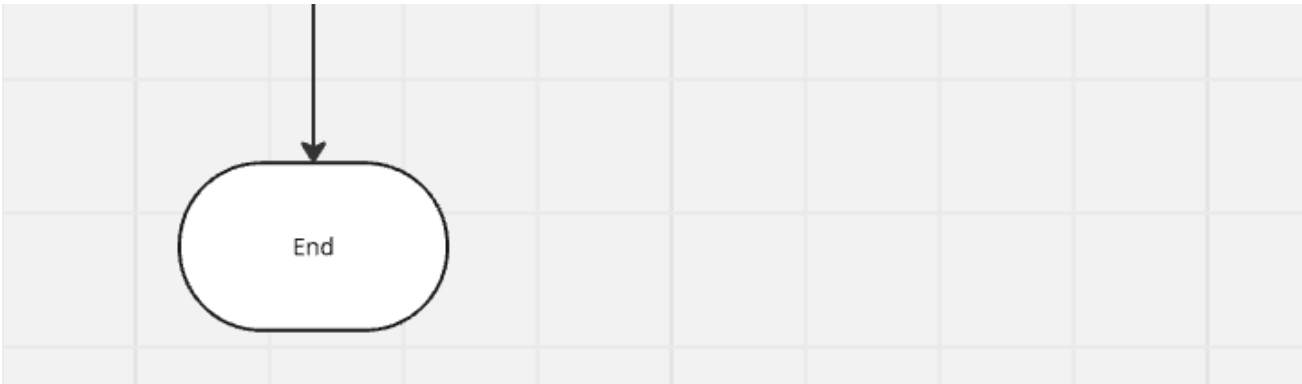
These flowcharts show how user input is handled and what actions the program takes.

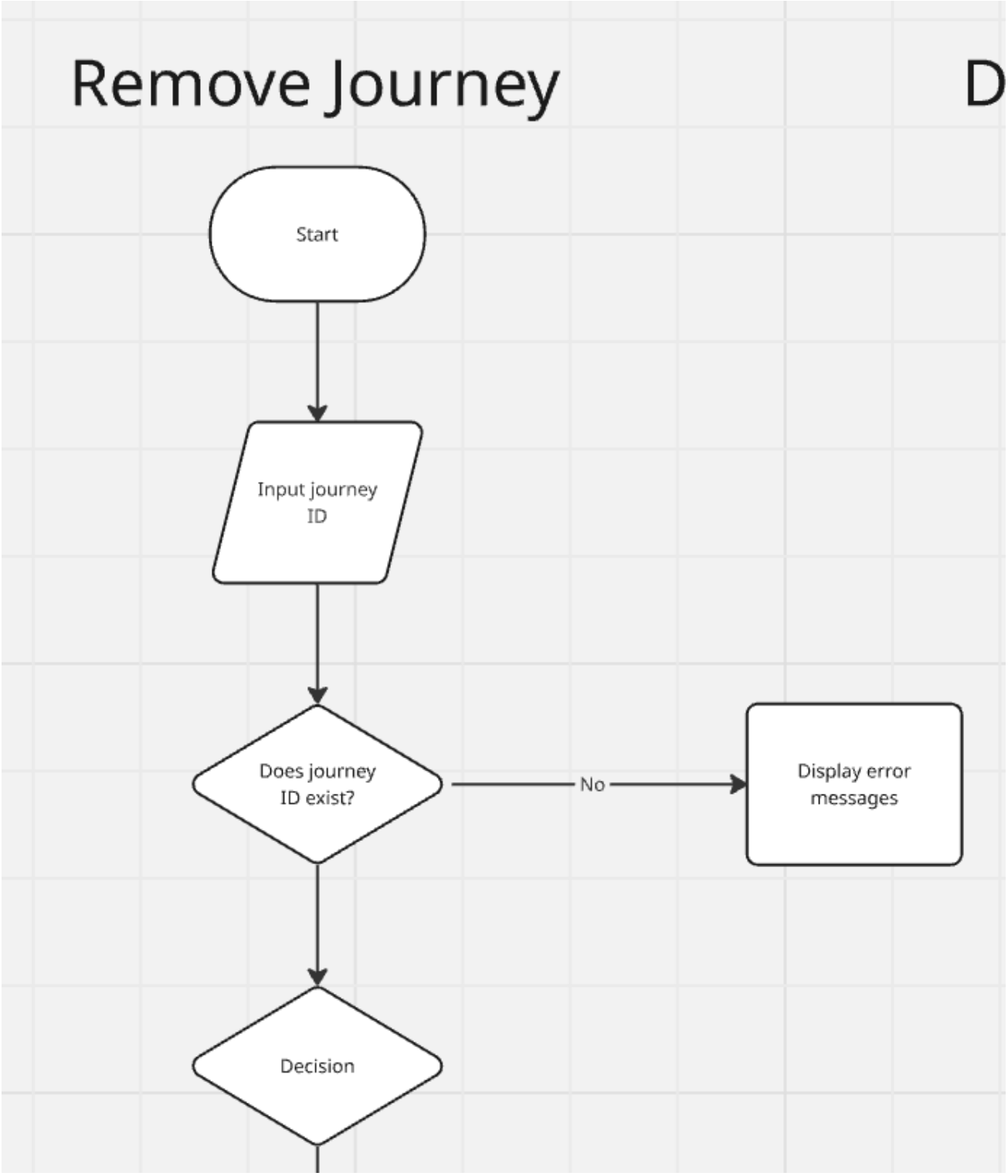

Start
Initialise journey list and totals
Display menu options
input user menu
Add journey
Remove journey

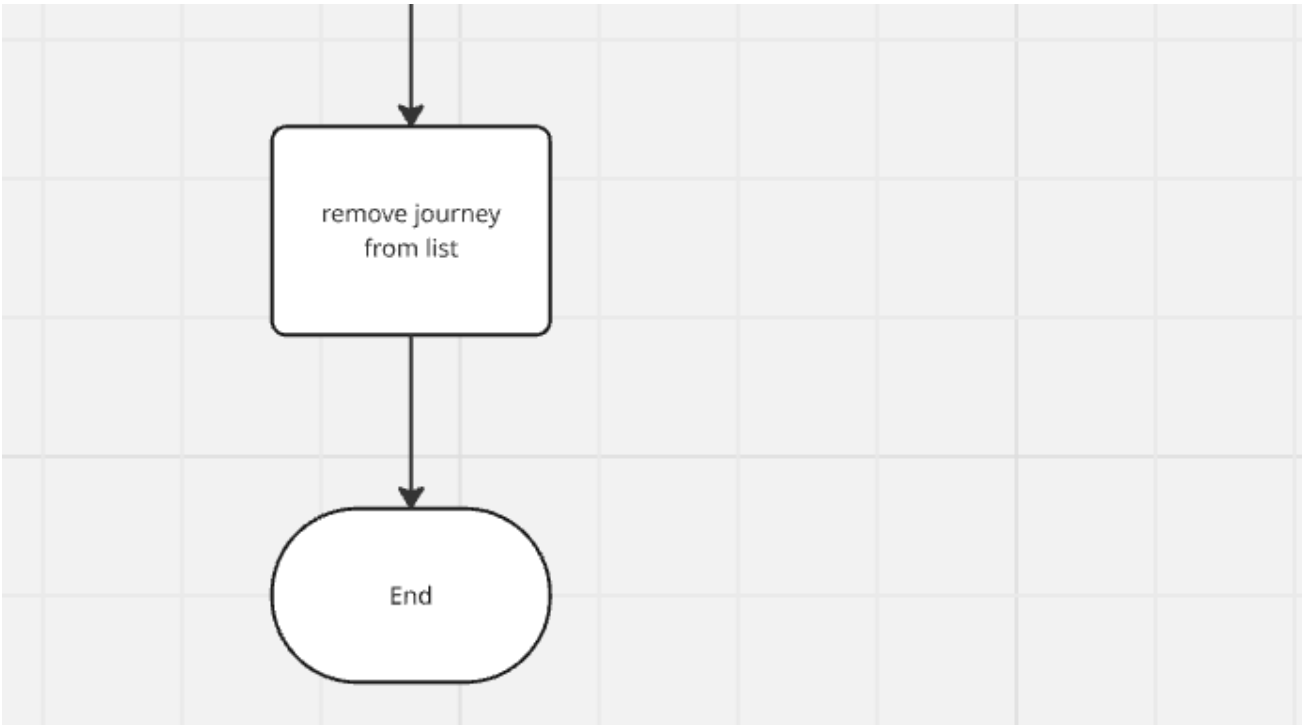
- 
Daily summary
End
-
-
-

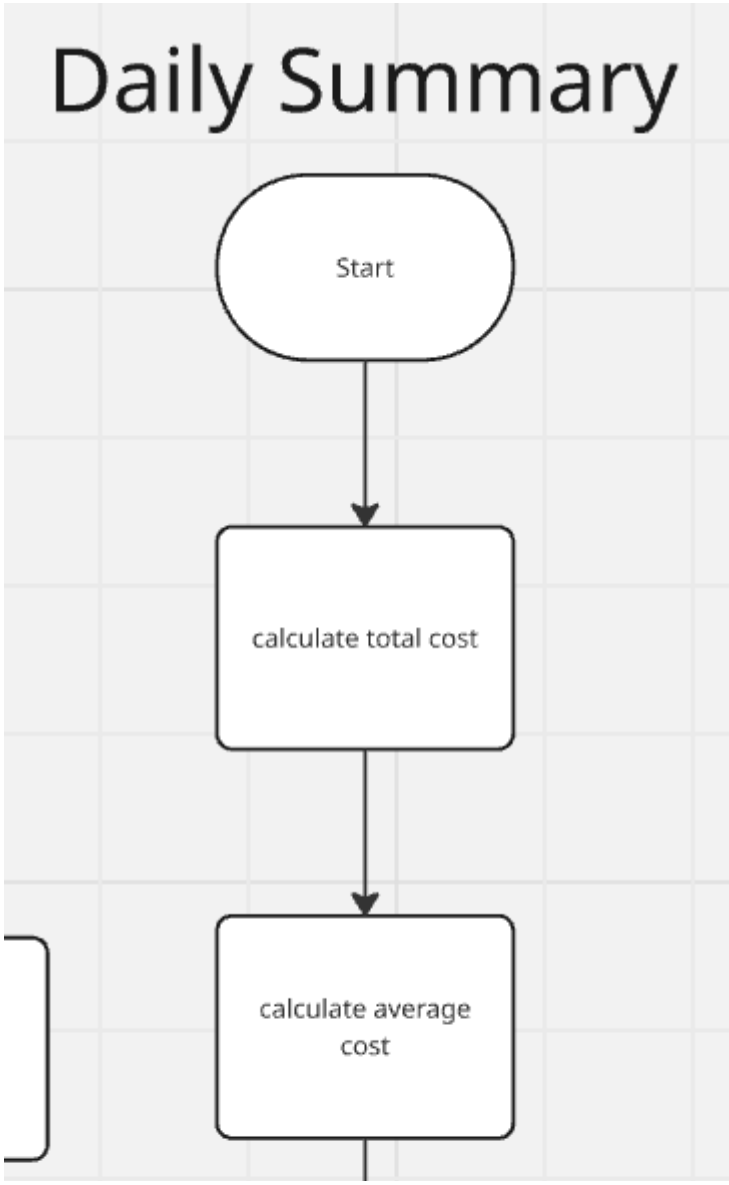

Add Journey
Start

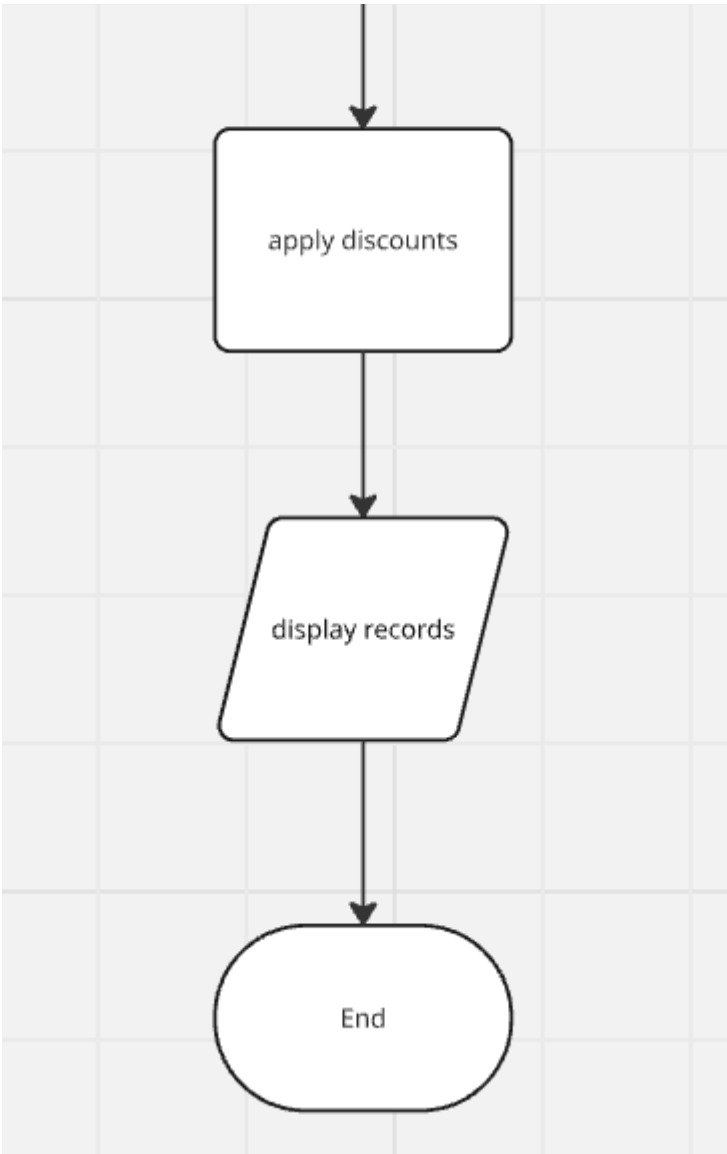

Add journey
Input date, zone, passenger, type, time
check inputs valid
No
display error message
calculate fare
Generate journey ID, Store journey in list
display confirmation message

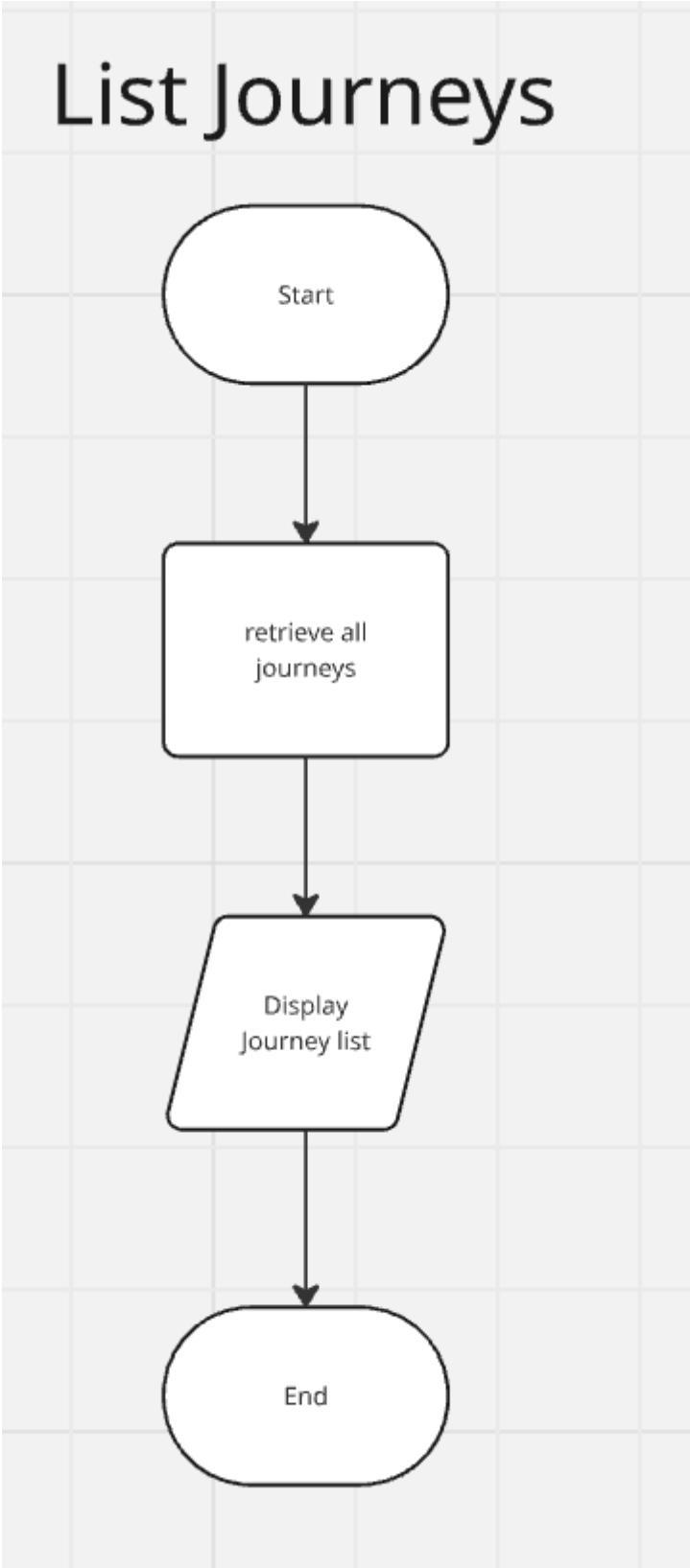
- 
End

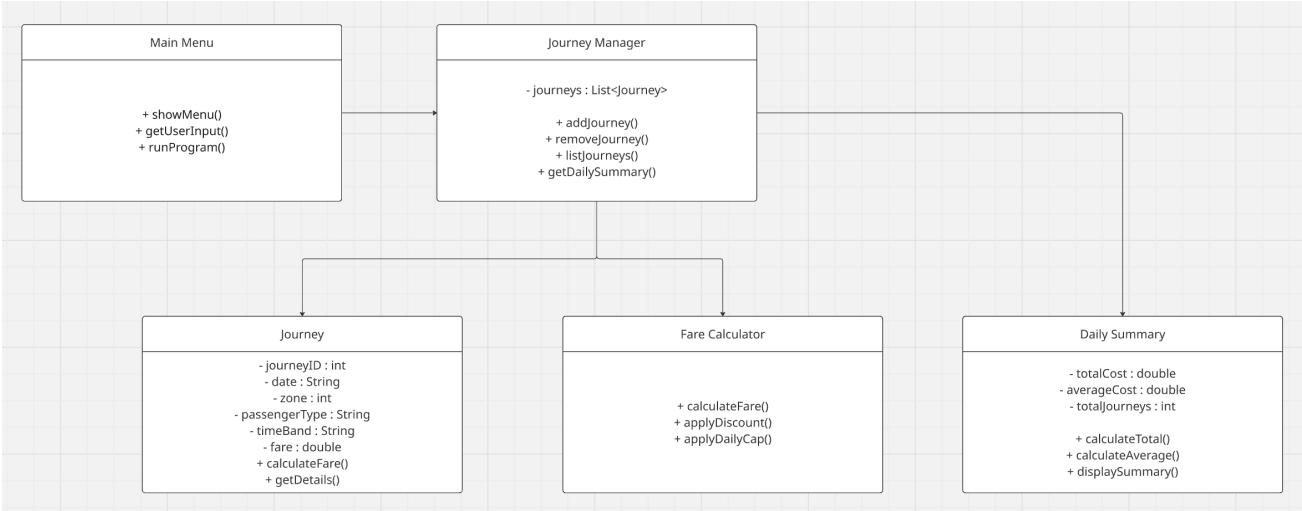

Remove Journey
D
Start
Input journey ID
Does journey ID exist?
No
Display error messages
Decision

- 
remove journey from list
End


Daily Summary
Start
calculate total cost
calculate average cost

- 
apply discounts
display records
End

- 
List Journeys
Start
retrieve all journeys
Display Journey list
End

- 

Research

Research existing programs that solve a similar problem. The program does not have to be written in java or object orientated in nature - just solve a similar type of problem.

Use the strucutre below to capture your evidence:

Name of program: Moovit

Reference (link): https://moovitapp.com/

What it does well :What it does well:

- Provides real-time public transport journey planning.
- Allows users to view routes based on current location.
- Supports multiple transport types.

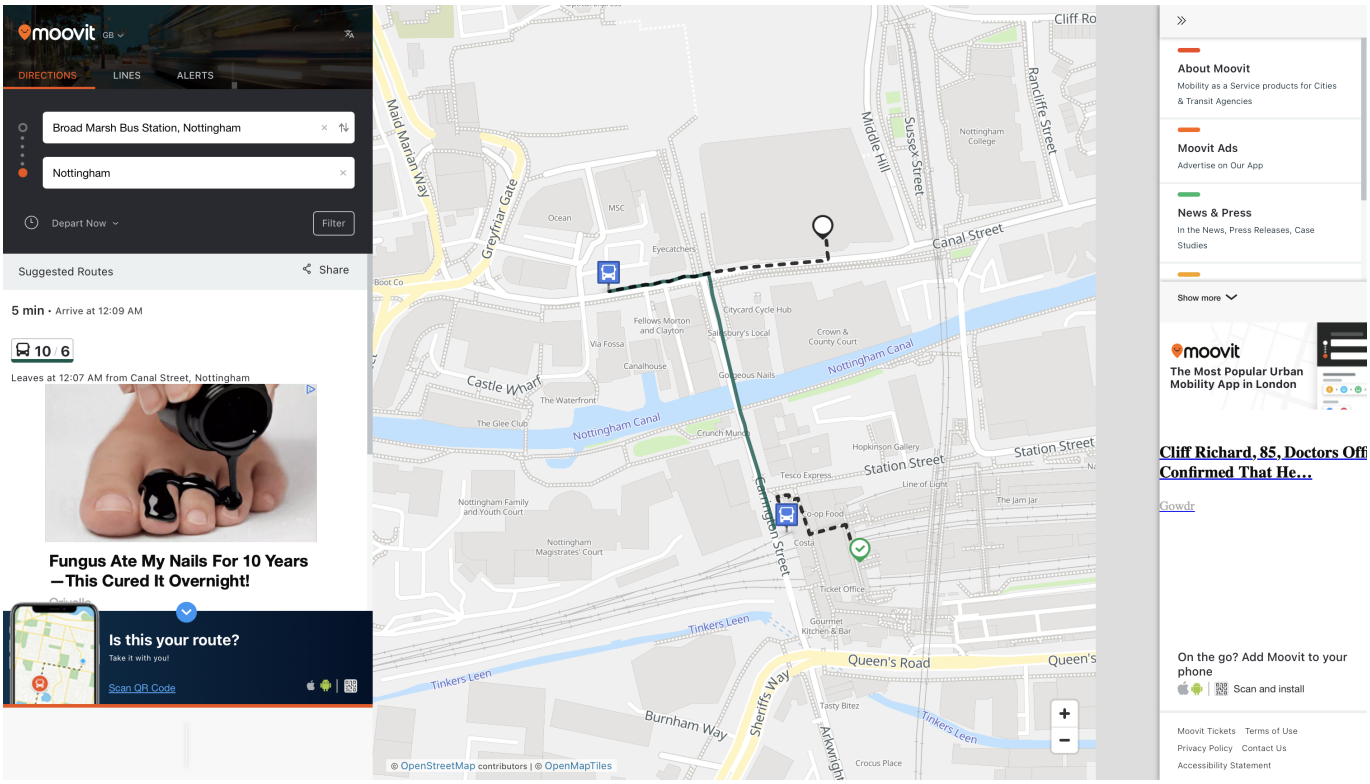
What it does poorly :

- Does not calculate journey fare costs.
- Does not offer daily summary information.

Key design ideas you could reuse (e.g., layout, navigation, input/output, program structure):

- Real-time journey listing for user input/output.

Screenshot (showing the interface/outp



Updated Gantt Chart

	A	B	C
1	**Task**	**Start Date**	**End Date**
2	Milestone 1 – Requirements analysis	22/01/2026	22/01/2026
3	Milestone 1 – IPO table and planning	24/01/2026	24/01/2026
4	Milestone 1 – Review and documentation	25/01/2026	25/01/2026
5	Milestone 2 – Algorithm design (flowcharts)	29/01/2026	29/01/2026
6	Milestone 2 – Class diagram design	30/01/2026	30/01/2026
7	Milestone 2 – Research similar application (Moovit)	31/01/2026	31/01/2026
8	Milestone 2 – Report writing and screenshots	31/01/2026	01/02/2026
9	Milestone 2 – Final review and submission	01/02/2026	01/02/2026

Diary Entries

29/01/2026 – Diary Entry 4

I started working on the algorithm design for the CityRide Lite program. I created flowcharts for the main features such as Main Menu, Add Journey, Remove Journey, Daily Summary and List

Journeys. This helped me understand how the program should work step by step and how user input is processed.

30/01/2026 – Diary Entry 5

I designed the class diagram for the program. I identified the main classes such as Journey, Journey Manager, Fare Calculator and Daily Summary. I added attributes and methods to each class to show how the system is structured. This made the object-oriented design of the program clearer.

31/01/2026 – Diary Entry 6

I researched an existing transport application (Moovit) that solves a similar problem. I analysed what the program does well and what it does poorly. I also identified design ideas that could be reused in my own program, such as clear navigation and simple input and output design. I updated the Gantt chart and added all diagrams and screenshots to the report.